

Coordinated Motion Control of Moving Gantry Stages for Precision Applications Based on an Observer-Augmented Composite Controller

K. K. Tan, S. Y. Lim, S. Huang, H. F. Dou, and T. S. Giam

Abstract—High-precision operation in gantry systems is required to meet higher demands on positioning accuracy in a wide range of applications such as circuit assembly, precision metrology, and wafer stepping. In this paper, the experimental study of an observer-augmented composite control scheme for coordinated motion control of moving gantry stages for precision applications is presented. The scheme will be implemented and compared to other control schemes currently available for this purpose, including a master-slave and a set-point coordinated scheme. Experimental results will illustrate the enhanced performance of the fully coordinated control scheme with respect to tight trajectory tracking purposes.

Index Terms—Disturbance observer, fully coordinated control, master-slave control, moving gantry stages, set-point coordinated control.

I. INTRODUCTION

AMONG the various configurations of long travel and high-precision Cartesian robotic systems, one of the most popular is the *H*-type which is more commonly known as the moving gantry stage. In this configuration, two motors which are mounted on two parallel slides move a gantry simultaneously in tandem. This gantry configuration has been in use for large overhead travelling cranes in ports, rolling mills and flying shear. When positioning precision is of the primary concern, direct drive linear motors are usually used and fitted with aerostatic bearings for optimum performance. When used particularly with permanent magnet linear motors (PMLM), the moving gantry stage is designed to provide a high-speed high-accuracy *X*, *Y*, and *Z* motion to facilitate automated processes in flat panel display, printed circuit board manufacturing, precision metrology, and circuit assembly where high-part placement accuracy for overhead access is necessary. The stage is equipped with a high-power density due to the dual drives, and it can yield high-speed motion with no significant lateral offset when the two drives are appropriately coordinated and synchronised in motion. In certain applications such as in wafer steppers, the dual drives can also be used to produce a small “theta” rotary motion, without any additional rotary actuators. The application domain of the moving gantry stage is rapidly expanding because of the increasingly stringent

requirements arising from developments in precision engineering and nanotechnology [1], [2]. To date, there is not one precision machine manufacturer who does not provide for a moving gantry stage in one form or another.

The main challenges to address in order to harness the full potential of this configuration of cartesian stages are mainly in the control system. In addition to precision motion control of the individual motors to track the desired motion trajectories, efficient synchronization among them is crucially important to minimize the positional offsets between the two axes. This offset arises due to different drive and motor characteristics, nonuniform load distribution of the gantry and attached end-effectors, and time-varying thermo-mechanical properties. In particular, disturbances in the form of dynamical load changes which can be fairly asymmetrical in nature, have to be adequately addressed. To the best of our knowledge, there has seemed to be little effort devoted to address the motion control aspects of a moving gantry robotic system on a highly precise scale, even though these are fast becoming important enabling technologies to facilitate the fulfillment of high-accuracy processes.

In this paper, current control approaches in commercial motion control platforms are first surveyed and presented. The performance of these schemes and their limitations will be investigated and compared in this paper through experiments. The first scheme assigns one motor as the master and the other as the slave. The command signal is only transmitted to the master, and the slave simply tracks the motion of the master. The second and more recent scheme addresses explicitly the possibly different dynamics of the two motors. An independent control loop under a common command signal is set up for each motor, in a configuration similar to supervisory control. However, similar to the master-slave scheme, there is still no feedback of interaxis motion offset for a truly coordinated control.

A novel scheme involving an observer-augmented composite controller is proposed and developed in the paper to overcome the deficiencies of the aforementioned ones. It possesses several important and useful features. First, it incorporates a feedforward component to facilitate a high-speed response. The feedforward component addresses model-based characteristics relating to the servo motors, including specific friction, ripples and possibly other torque-impeding characteristics. Second, an optimal PID feedback controller is designed based on a linear quadratic regulator (LQR) approach [3]. This PID feedback control is intended to provide optimal command response and stability properties. Where additional state variable measurements are available, full state feedback controllers may be designed along similar design rules to achieve additional enhancement. Third, a disturbance observer is augmented to the composite

Manuscript received September 14, 2001; revised September 3, 2002. Manuscript received in final form March 30, 2004. Recommended by Associate Editor F. Ghorbel.

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Digital Object Identifier 10.1109/TCST.2004.833608

control structure to provide a fast response to load disturbances and other exogenous signals acting asymmetrically on the two motors. This feature is useful since load disturbances are major factors affecting the control performance, especially when the motors jointly carry a dynamical and asymmetrical load such as an additional servo system running across the gantry. The disturbance observer acts as a soft sensor to derive the equivalent disturbance signals based only on prevailing output signals and available system models. The disturbances can then be efficiently and automatically corrected for. Experimental results are duly provided in this paper to illustrate the relative performance of the control schemes and their competitive advantages.

The paper is organized as follows. Section II presents a mathematical model of servo systems which will be used in many of the control components. Section III covers the principles and details of the current motion control schemes for moving gantry stages. The development of the proposed observer-augmented composite controller is covered in Section IV. Experimental results on two configurations of *XY* table under the various scenarios are presented in Section V.

II. SYSTEM DESCRIPTION

Different kinds of motors may be used in the construction of a gantry motion system, depending on the precision and cost requirements. For high-precision applications, dc PMLMs are gaining popularity due to their mechanical simplicity and direct drive features. Whichever is the choice of motors, servo dynamics can be viewed as essentially comprising of two components: a dominantly linear model, and an uncertain and nonlinear remnant which nonetheless must be considered in the design of the control system if high-precision motion control is to be efficiently realized.

In the dominant linear model, the mechanical and electrical dynamics of a servo motor can be expressed as follows:

$$M\ddot{x} + D\dot{x} + F_{\text{load}} = F_m \quad (1)$$

$$K_e\dot{x} + L_a \frac{dI_a}{dt} + R_a I_a = u \quad (2)$$

$$F_m = K_t I_a \quad (3)$$

where x denotes position; M , D , F_m , F_{load} denote the mechanical parameters: inertia, viscosity constant, generated force, and load force, respectively, u , I_a , R_a , L_a denote the electrical parameters: input dc voltage, armature current, armature resistance, and armature inductance, respectively; K_t denotes an electrical-mechanical energy conversion constant; K_e is the back electromotive force (EMF) constant of the motor.

Since the electrical time constant is typically much smaller than the mechanical one, the delay due to electrical transient response may be ignored, giving the following simplified model:

$$\ddot{x} = -\frac{K_1}{M}\dot{x} + \frac{K_2}{M}u - \frac{1}{M}F_{\text{load}} \quad (4)$$

where

$$K_1 = \frac{K_e K_t + R_a D}{R_a} K_2 = \frac{K_t}{R_a}. \quad (5)$$

Clearly, this is a second-order linear dynamical model.

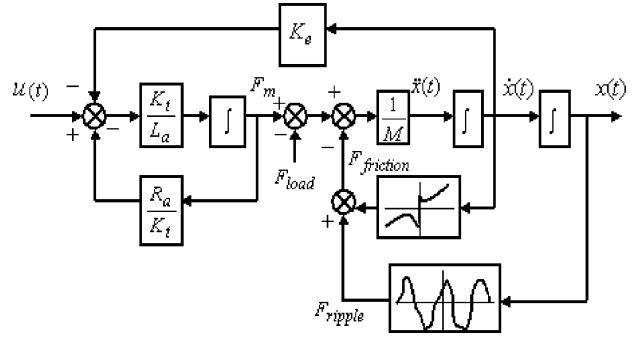


Fig. 1. Model of servo motor.

The dominant linear model has not included extraneous nonlinear effects which may be present in the physical structure. Among them, prominent nonlinear effect associated with servo motors are due to frictional and ripple forces. Fig. 1 depicts a block diagram model of a servo motor, including explicitly the various exogenous disturbance signals present.

A. Friction

Friction is inevitably present in nearly all moving mechanisms, and it is one major obstacle to achieving precise motion control. Several characteristic properties of friction have been observed, which can be broken down into two categories: static and dynamic. The static characteristics of friction, including the stiction friction, the kinetic force, the viscous force, and the Stribeck effect, are functions of steady-state velocity. The dynamic phenomena include presliding displacement, varying breakaway force, and frictional lag. Many empirical friction models have been developed which attempt to capture specific components of observed friction behavior, but generally it is acknowledged that a precise and accurate friction model is difficult to be obtained in an explicit form, especially for the dynamical component. For many purposes, however, the Tustin model has proven to be useful and it has been validated adequately in many successful applications [4]. The Tustin model may be written as

$$F_{\text{friction}} = [F_c + (F_s - F_c)e^{-(|\dot{x}/\dot{x}_s|)^\delta} + F_v|\dot{x}|]\text{sgn}(\dot{x}) \quad (6)$$

where F_s denotes static friction, F_c denotes the minimum value of Coulomb friction, $\dot{x}_s$ and F_v are lubricant and load parameters, and δ is an additional empirical parameter. Fig. 2 graphically illustrates this friction model.

Considering these nonlinear effects, the PMLM dynamics may be described by

$$\ddot{x} = -\frac{K_1}{M}\dot{x} + \frac{K_2}{M}u - \frac{1}{M}(F_{\text{load}} + F_{\text{ripple}} + F_{\text{friction}}). \quad (7)$$

The effects of friction can be greatly reduced using high-quality bearings such as aerostatic or magnetic bearings.

B. Force Ripples

The phenomenon of force ripples is especially evident in dc permanent magnet motors such as PMLMs [5]. The two primary components of the force ripple are the cogging (or detent) force and the reluctance force. The cogging force arises as a result

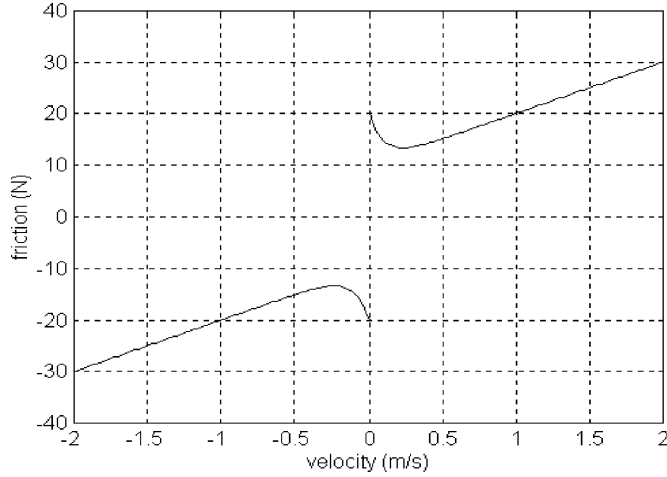


Fig. 2. Tustin friction model.

of the mutual attraction between the magnets and iron cores of the stator. This force exists even in the absence of any winding current and it exhibits a periodic relationship with respect to the position of the translator relative to the magnets. Cogging manifests itself by the tendency of the translator to align in a number of preferred positions regardless of excitation states. There are two potential causes of the periodic cogging force in PMLMs, resulting from the slotting and the finite length of the iron-core translator. The reluctance force is due to the variation of the self-inductance of the windings with respect to the relative position between the translator and the magnets. Thus, the reluctance force also has a periodic relationship with the translator-magnet position.

Collectively, the cogging and reluctant force constitute the overall force ripple phenomenon. Even when the motor is not powered, force ripples are clearly existent when the translator is moved along the guideway. There are discrete points where minimum/maximum resistance is experienced. At lower velocity, the rippling effects are more fully evident due to the lower momentum available to overcome the magnetic resistance.

A first-order model for the force ripple for a PMLM can be described as a periodic sinusoidal type signal

$$F_{\text{ripple}}(x) = A(x) \sin(\omega x + \phi). \quad (8)$$

Higher harmonics of the ripple may be included in higher order models.

A common nonlinear function $F_1^*(x, \dot{x})$ may be used to represent the nonlinear dynamical effects due to friction, force ripple, and other unaccounted dynamics collectively. The servo system (7) can, thus, be alternatively described by

$$\ddot{x} = -\frac{K_1}{M}\dot{x} + \frac{K_2}{M}u - \frac{1}{M}F_{\text{load}} + F_1^*(x, \dot{x}). \quad (9)$$

Let

$$\frac{K_2}{M}f(x, \dot{x}) = -\frac{1}{M}F_{\text{load}} + F_1^*(x, \dot{x}). \quad (10)$$

It follows that

$$\ddot{x} = -\frac{K_1}{M}\dot{x} + \frac{K_2}{M}u + \frac{K_2}{M}f(x, \dot{x}). \quad (11)$$

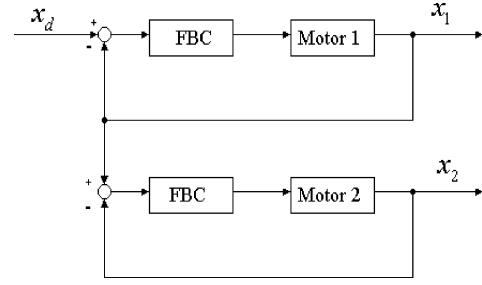


Fig. 3. Block diagram of the master-slave control scheme.

With the tracking error e defined as

$$e = x_d - x$$

(11) may be expressed as

$$\ddot{e} = -\frac{K_1}{M}\dot{e} - \frac{K_2}{M}e - \frac{K_2}{M}f(x, \dot{x}) + \frac{K_2}{M}\left(\frac{M}{K_2}\ddot{x}_d + \frac{K_1}{K_2}\dot{x}_d\right). \quad (12)$$

Since

$$\frac{d}{dt} \int_0^t e(t)dt = e \quad (13)$$

the system state variables are assigned as $x_1 = \int_0^t e(t)dt$, $x_2 = e$ and $x_3 = \dot{e}$. Denoting $X = [x_1 \ x_2 \ x_3]^T$, (12) can then be put into the equivalent state-space form

$$\dot{X} = AX + Bu + Bf(x, \dot{x}) + B\left(-\frac{M}{K_2}\ddot{x}_d - \frac{K_1}{K_2}\dot{x}_d\right) \quad (14)$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -\frac{K_1}{M} \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ -\frac{K_2}{M} \end{bmatrix}. \quad (15)$$

III. SURVEY OF CONTROL SCHEMES FOR GANTRY MOTION SYSTEMS

Two control schemes commonly used in current industrial control systems for the control of moving gantry stages will be duly described in the following sections.

A. Classical Master-Slave Approach

In the so-called “classical master-slave” approach [6], one motor is chosen as the master motor of the pair. This master motor executes directly the desired trajectories. The encoder of the master motor is also used as the master encoder for the parallel slave motor. Thus, in essence, the slave motor simply follow the motion of the master motor. Fig. 3 depicts a block diagram of the master-slave control scheme.

This method is relatively very simple and it works just as if a single motor were driving the motion. It is an effective method to use also during jogging and homing moves; the master motor is commanded to jog or home, and the slave motor simply follows along. However, the performance achievable in terms of trajectory tracking can be rather limited, due to the fact that the actual trajectory of the master motor acts as the commanded trajectory of the slave motor [6]. Clearly, the actual trajectory of the master motor can never be as smooth and accurate as the commanded trajectory and if this now becomes the commanded trajectory of

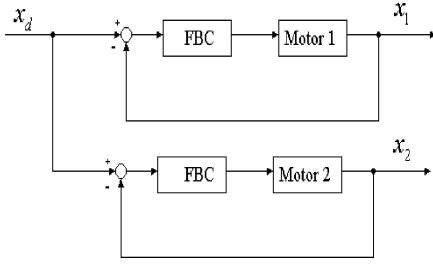


Fig. 4. Block diagram of the set-point coordinated control scheme.

the slave motor, the actual trajectory of the slave motor will deviate even more from the desired one. This is particularly true on gantry systems where both motors have similar resonant frequencies. Oscillations in the master motor will inevitably be fed into the slave motor and get amplified significantly. In addition, when the slave motor encounters a disturbance, the master will not be able to know and appropriately address that.

B. Set-Point Coordinated Control

In the “set-point coordinated control” method [6], two motors are assigned to the same axis in the same coordinate system. A supervisory motion program drives the gantry along the axis through the two motors simultaneously. The motors use an identical desired trajectory commanded for the motion. Each servo loop of the two motors then has the responsibility of keeping the actual trajectory as closely as possible to the commanded trajectory, since each of them has its own individual servo loop. Presuming all motors have tight servo loops, this method provides a tight and smooth link between the motors for the gantry. Fig. 4 provides a block diagram of the supervisory control scheme.

Since the deficiencies of one motor in following the commanded trajectory do not directly affect the other motor, this scheme is in general superior to using a master–slave technique. The main problem with this form of control is due to differences in the dynamics of the two motors or in their load, in which case there may be a rather considerable difference in the positions of the motors.

IV. FULLY COORDINATED CONTROL BASED ON AN OBSERVER-AUGMENTED COMPOSITE CONTROL SYSTEM

The overall control structure is shown in Fig. 5.

In what follows, the purpose and design of each component depicted in Fig. 5 will be elaborated.

A. Feedforward Control

The design of the feedforward control component is straightforward. From (14), the term of $B(-(M/K_2)\ddot{x}_d - (K_1/K_2)\dot{x}_d)$ may be neutralized using a feedforward control term in the control signal. The feedforward control is, thus, designed as

$$u_{FF,1}(t) = \frac{M}{K_2}\ddot{x}_d + \frac{K_1}{K_2}\dot{x}_d. \quad (16)$$

Clearly, the reference position trajectory must be continuous and twice differentiable, otherwise a precompensator to filter the reference signal will be necessary. The only parameters required for the design of the feedforward control are the parameters of the second-order linear model.

Additional feedforward terms may be included for direct compensation of the nonlinear effects, if the appropriate models are available. A friction feedforward compensator can be designed as

$$u_{FF,2}(t) = \frac{1}{K_2}[F_c + (F_s - F_c)e^{-(|\dot{x}_d/\dot{x}_s|)^\delta} + F_v|\dot{x}_d|]sgn(\dot{x}_d).$$

In [7], an efficient way of friction modeling using relay feedback is proposed where a simple friction model (incorporating coulomb and viscous friction components) can be obtained automatically. This is achieved by using a relay in the feedback loop to generate sustained small oscillations from which the parameters of the friction model may be estimated. This model can be used to construct an additional feedforward control signal, based only on the reference trajectories.

In the same way, a ripple feedforward precompensator can be installed if the force ripples present are significant

$$u_{FF,3} = \frac{1}{K_2}F_{\text{ripple}}(x_d) = \frac{1}{K_2}A(x_d)\sin(\omega x_d + \phi).$$

Characteristic of all feedforward control schemes, the performance is critically dependent on the accuracy of the model parameters. Therefore, feedforward is usually augmented with suitable feedback control schemes.

B. PID Feedback Control

In spite of the advances in mathematical control theory over the last 50 years, industrial servo control loops are still essentially based on the three-term PID controller. The main reason is due to the widespread field acceptance of this simple controller which has been effective and reliable in most situations when adequately tuned. More complex advanced controllers have fared less favorably under practical conditions, despite the higher costs associated with implementation and the higher demands in control tuning. It is very difficult for operators unfamiliar with advanced control to adjust the control parameters. Given these uncertainties, there is little surprise that PID controllers continue to be manufactured by the hundred thousands yearly and still increasing. In the composite control system, PID is used as the feedback control term. While the simplicity in a PID structure is appealing, it is also often proclaimed as the reason for poor control performance whenever it occurs. In this design, advanced optimum control theory is applied to tune PID control gains. The PID feedback controller is designed using the LQR technique for optimal and robust performance of the nominal system. The feedforward plus feedback configuration is often also referred to as a two degree-of-freedom (2-DOF) control.

The nominal portion of the system (without uncertainty) is given by

$$\dot{x}(t) = Ax(t) + Bu(t) \quad (17)$$

where

$$u_{\text{PID}} = Kx = kx_1 + k_{d1}x_2 + k_{d2}x_3. \quad (18)$$

This is a PID control structure which utilizes a full-state feedback.

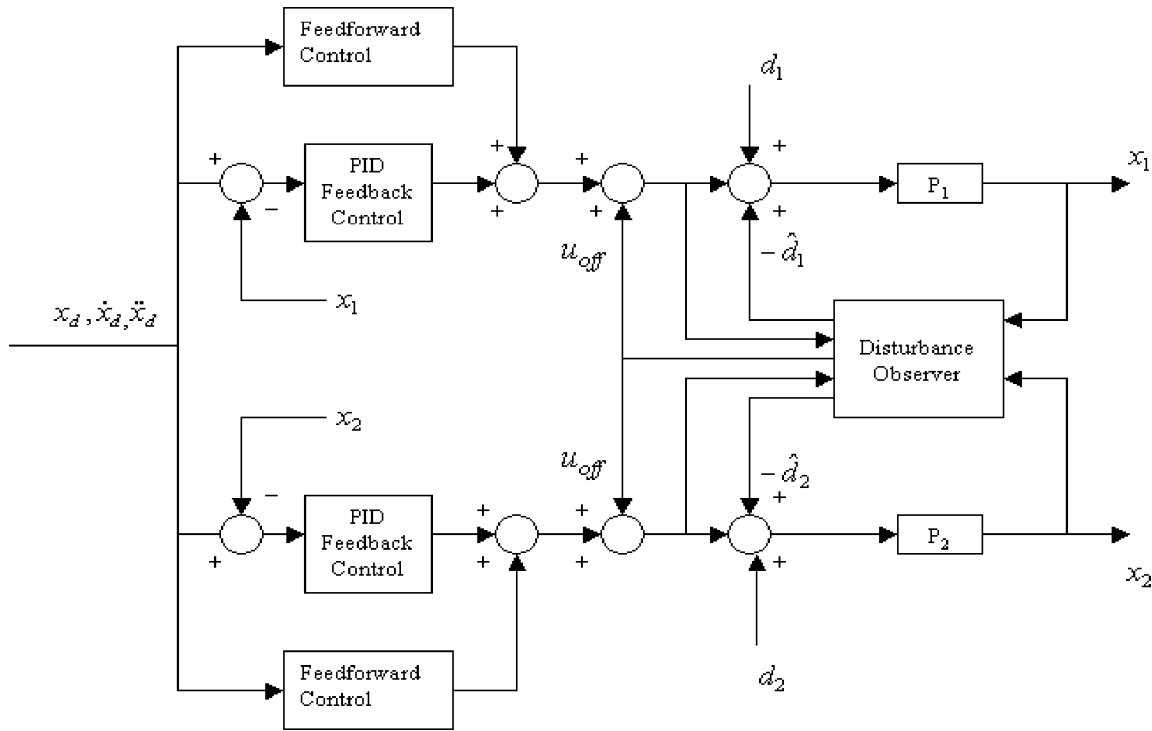


Fig. 5. Overall structure of control system.

The optimal PID control parameters are obtained using the LQR technique that is well known in modern optimal control theory and it has been widely used in many applications. It has a very nice robustness property, i.e., if the process is of single-input and single-output, then the control system has at least a phase margin of 60° and a gain margin of infinity. Under mild assumptions, the resultant closed-loop system is always stable. This attractive property appeals to the practitioners. Thus, the LQR theory has received considerable attention since 1950s.

The PID control is given by

$$u_{\text{PID}} = -(r_0 + 1)B^T P x(t) \quad (19)$$

where P is the positive definite solution of the Riccati equation

$$A^T P + P A - P B R^{-1} B^T P + Q = 0 \quad (20)$$

R relates to the control weight, and $Q = H^T H$ where H relates to the states weighting parameters in the usual manner. Note that r_0 is independent of P and it is introduced to weigh the relative importance between control effort and control errors. Note for this feedback control, the only parameters required are the parameters of the second-order model and a user-specified error weight r_0 .

C. Disturbance Observer

The achievable performance of a precision gantry system is unavoidably and very significantly limited by the amount of disturbances present, and the uniformity of the distribution among the motors [8]. These disturbances may arise due to load changes, system parameter perturbation owing to prolonged usage, measurement noise, high frequencies generated from the amplifiers (especially when a *pulse-width modulated* (PWM) amplifier is used), or inherent nonlinear dynamics such as the

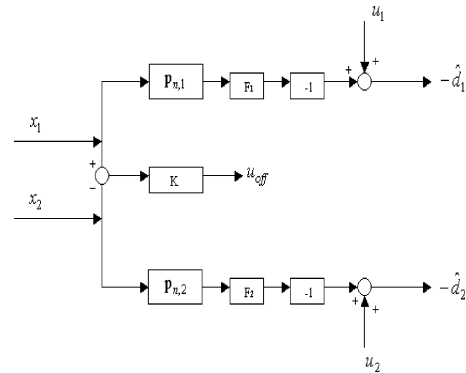


Fig. 6. Block diagram of the disturbance observer.

force ripples and frictional forces mentioned. Incorporating a higher resolution in the measurement system via the use of high interpolation electronics on the encoder signals can only achieve improvement in positioning accuracy to a limited extent. Thereafter, the amount of disturbances present will ultimately determine the achievable performance. In this subsection, this important issue of disturbance compensation for precision motion control systems will be addressed.

Fig. 6 shows the block diagram of the “disturbance observer” part of the proposed control system which uses an estimate of the actual disturbance, deduced from a disturbance observer, to compensate for the disturbances. x_i , u_i , d_i and $\hat{d}_i$ denote the position signal, control signal, actual and observed disturbance signal associated with the i axis. u_{off} represents a constituent control signal to correct for the interaxis offset. This can be the output from a P or PI control. Where the measurement noise is tolerable, an additional derivative action, possibly in conjunction with a filter, may also be installed to provide a derivative

TABLE I
PARAMETERS OF MT22G2 dc SERVO MOTOR

Contents	Units	MT22G2-10
Torque Constant (K_t)	N/Amp	0.10
Armature Resistance Less Brushes (R_a)	Ohms	0.63
Voltage Constant EMF (K_e)	volt/rad/sec	0.10
Armature Inductance (L_a)	mh	2.1
Max. velocity	RPM	5000

offset correction. The disturbance observer estimates the disturbance based on the output x_i and the control signal u_i . P_i denotes the actual system. $P_{n,i}$ denotes the nominal system which can be generally described by

$$P_{n,i} = \frac{a_{0,i}}{s^l(s^{m-l} + a_{1,i}s^{m-l-1} + \dots + a_{m-l,i})} \quad (21)$$

where $P_{n,i}$ is a m th-order delay system and has l poles at the origin. In this paper, as mentioned, we will use a third-order model, i.e., $l = 1$, $m = 3$, according to (4)

$$P_{n,i} = \frac{a_{0,i}}{s(s^2 + a_{1,i}s + a_{2,i})}. \quad (22)$$

The disturbance observer incorporates the inverse of the nominal system and, thus, a low-pass filter F_i is required to make the disturbance observer proper and practically realizable. For our choice of a third-order model $P_{n,i}$, a suitable filter is

$$F_i(s) = \frac{f_{3,i}}{s^3 + f_{1,i}s^2 + f_{2,i}s + f_{3,i}}. \quad (23)$$

$f_{1,i}$, $f_{2,i}$, and $f_{3,i}$ can be adjusted, relative to the plant model $P_{n,i}$, to satisfy a satisfactory compromise between tracking and disturbance rejection [8]. The $f_{1,i}$, $f_{2,i}$, and $f_{3,i}$ have been adjusted to $f_{1,i} = 0.01$, $f_{2,i} = 1$, and $f_{3,i} = 9$.

V. EXPERIMENTS

Real-time experiments are carried out on two configurations of X - Y table. Both setups use a $2.5 \mu\text{m}$ resolution digital encoders installed on the x and y motors. The specifications for the motors are listed in Table I. The PID gain parameters used in the experiments are $P = 0.05$, $I = 1$, and $D = 0.0006$.

One table is configured to a moving gantry type, with two motors driving the load along the x direction. Fig. 7 shows a photograph of the table.

The other table is a more conventional one with one motor each along the x and y direction, the photograph of which is as shown in Fig. 8.

For the first configuration, the desired trajectory of the gantry for the experimental study is chosen to be a fifth-order polynomial as shown in Fig. 9. The control performance will be evaluated with respect to the tracking errors of the individual servo loops and the interloop motion offset.

For this second configuration, the control task in the experiment is to execute an XY diagonal as straightly and precisely as possible. Such requirements on precise diagonal motion are essential for the calibration of machine geometrical properties.

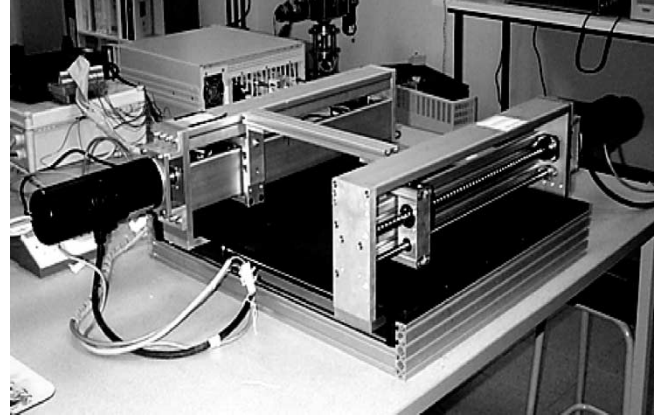


Fig. 7. XY table: configuration I.

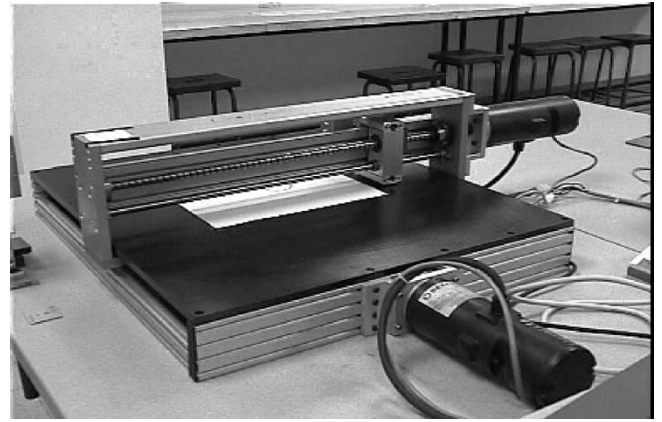


Fig. 8. XY table: configuration II.

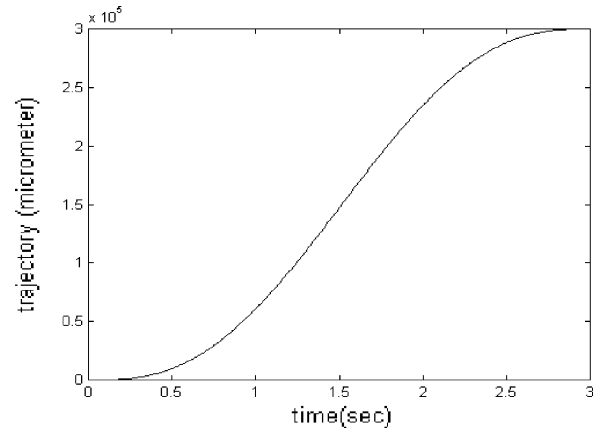


Fig. 9. Desired trajectory.

Clearly, in this application, a tight coordination between the X and Y motors is as important as the requirement for the moving gantry stage.

A. XY Table: Configuration I

Case-I: Master-Slave Control: Experimental results are shown in Fig. 10. A maximum tracking error of $15 \mu\text{m}$ is registered for the individual control loop with a position interaxis offset of as much as $16 \mu\text{m}$, reflecting the poor interaxis coordination in this control scheme. In high-precision applications,

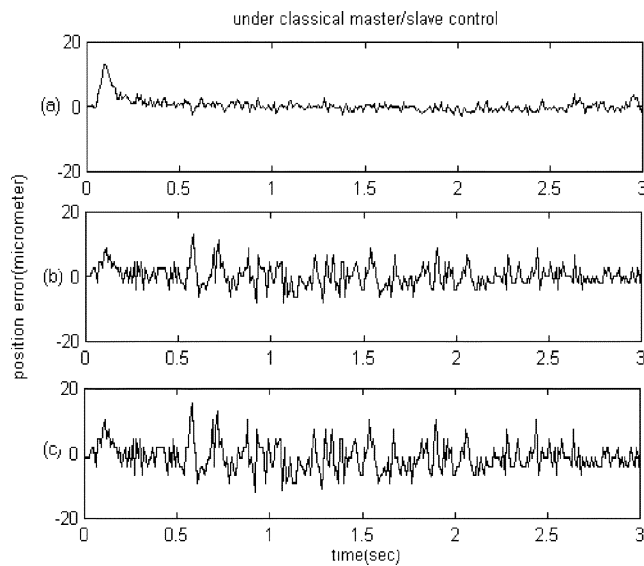


Fig. 10. (a) Tracking error for the x_1 axis under classical master–slave control. (b) Tracking error for the x_2 axis under classical master–slave control. (c) Positional interaxis offset under classical master–slave control.

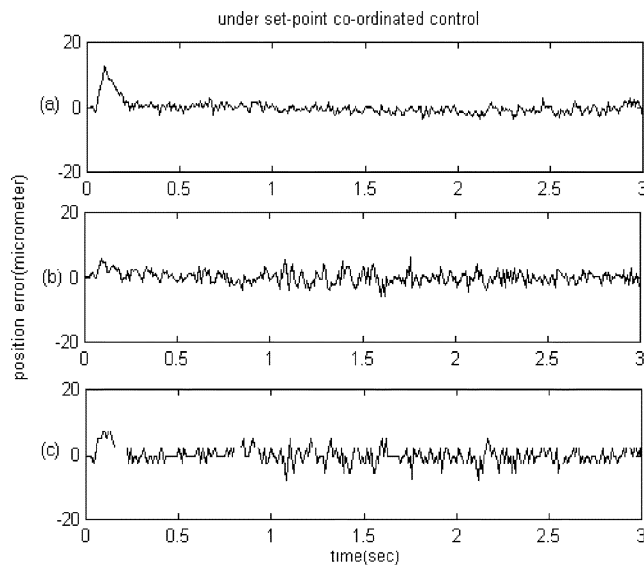


Fig. 11. (a) Tracking error for the x_1 axis under set-point coordinated control. (b) Tracking error for the x_2 axis under set-point coordinated control. (c) Positional interaxis offset under set-point coordinated control.

this offset may pose problems both in the mechanical alignment as well as in the achievements of the desired motion precision.

Case-2: Set-Point Coordinated Control: Under the set-point coordinated control scheme, each of the two axes has its own feedback control loop, but there is no feedback of the interaxis positional offset to either controller. The same PID controller in Case-1 is used in this case. Experimental results are shown in Fig. 11. A maximum tracking error of $13\text{ }\mu\text{m}$ is registered for the individual control loop. The interaxis position offset between the two axes x_1 and x_2 reaches $8\text{ }\mu\text{m}$, which is better than that under the master–slave control.

Case-3: Fully Coordinated Controller: In this case, apart from the individual control loop, the interaxis positional offset

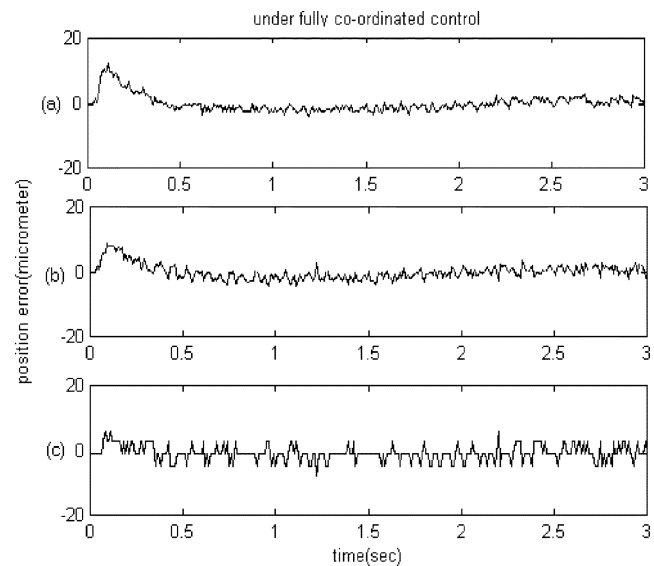


Fig. 12. (a) Tracking error for the x_1 axis under fully coordinated control. (b) Tracking error for the x_2 axis under fully coordinated control. (c) Positional interaxis offset under fully coordinated control.

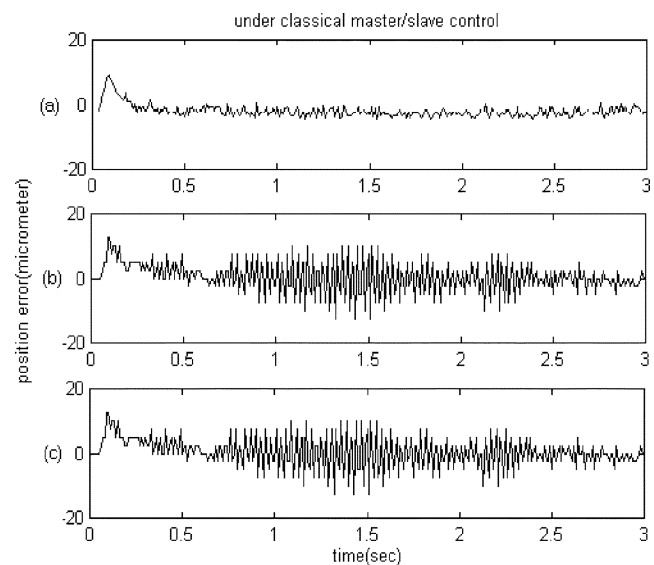


Fig. 13. (a) Tracking error for the x axis under classical master–slave control. (b) Tracking error for the y axis under classical master–slave control. (c) Positional interaxis offset under classical master–slave control.

is utilized to construct an additional control input for coordination purposes. Experimental results are shown in Fig. 12. A maximum tracking error of $12\text{ }\mu\text{m}$ is registered for the individual control loop. The maximum position interaxis offset between the two axes x_1 and x_2 is only $5\text{ }\mu\text{m}$, which is the best results among the three control schemes. It may be noted that the extent of improvement over the fully coordinated controller should depend on the amount and size of disturbance present. In the extreme case, if there is no disturbance, then the method offers little advantage. In this experiment, the gantry only carries a passive bridge. The disturbances come mainly in the form of the differences in the dynamics of the two motors. It is expected that a more significant improvement can be observed if

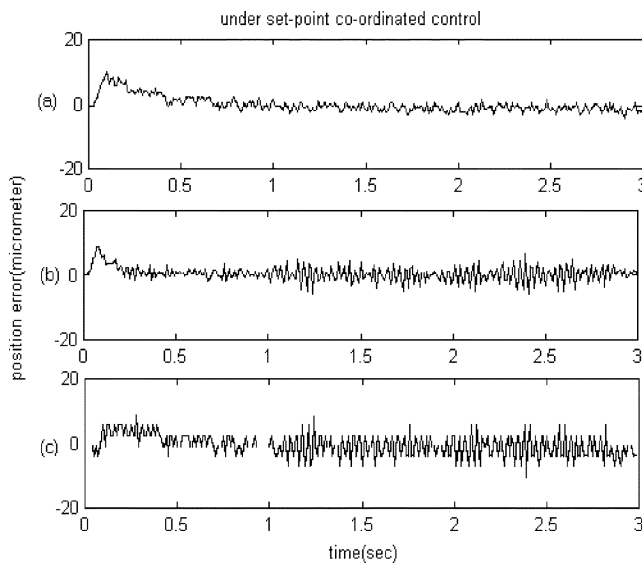


Fig. 14. (a) Tracking error for the x axis under set-point coordinated control. (b) Tracking error for the y axis under set-point coordinated control. (c) Positional interaxis offset under set-point coordinated control.

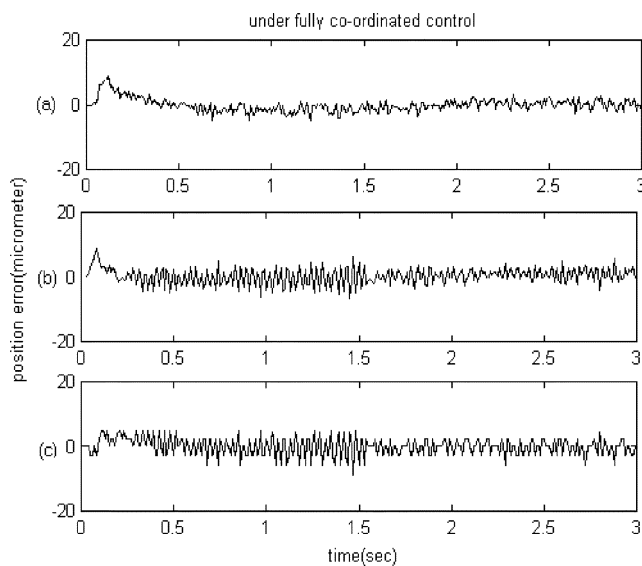


Fig. 15. (a) Tracking error for the x axis under fully coordinated control. (b) Tracking error for the y axis under fully coordinated control. (c) Positional interaxis offset under fully coordinated control.

the bridge is another slide transporting a tool which constitutes different loading on the motors.

B. XY Table: Configuration II

Case-1: Master-Slave Control: Experimental results are shown in Fig. 13. A maximum tracking error of $12\ \mu\text{m}$ is registered for the individual control loop.

Case-2: Set-Point Coordinated Control: Experimental results are shown in Fig. 14. A maximum tracking error of $8\ \mu\text{m}$ is registered for the individual control loop.

Case-3: Fully Coordinated Control: Experimental results are shown in Fig. 15. A maximum tracking error of $6\ \mu\text{m}$ is registered for the individual control loop, which is lower than the other two schemes. Fig. 16 shows a comparison of

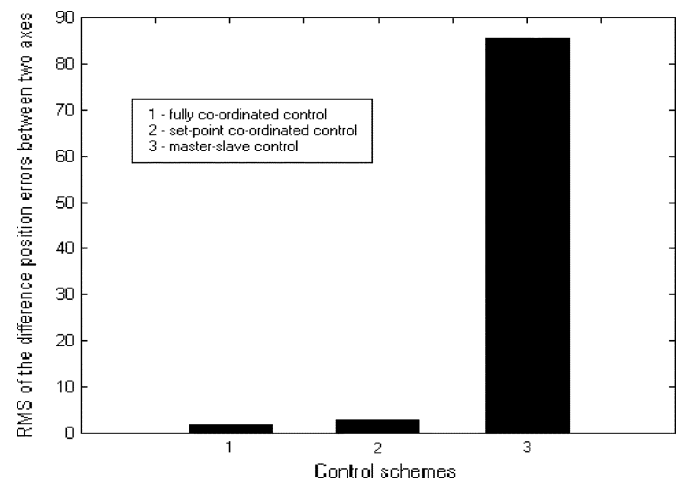


Fig. 16. Root mean square (rms) comparison of positional interaxis offset under different control schemes.

the interaxis offset arising from the use of each of the three control schemes. The interaxis positional offset with the fully coordinated control is also lower compared to the other two schemes, although the offset with the set-point coordinated scheme is also kept low, as there is no load disturbance (due to tool changing etc.) during the experiment.

VI. CONCLUSION

The coordinated motion control of moving gantry stages for precision applications has been addressed in this paper. A novel observer-augmented composite control scheme is presented. The scheme has been implemented and compared to control schemes currently available for this purpose, including a master-slave and a set-point coordinated scheme. Experimental results illustrate the enhanced performance of the fully coordinated control scheme with respect to tight trajectory tracking purposes.

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